

YEAR 9 SCIENCE
CHEMICAL SCIENCES
WORKSHEET - ACIDS, BASES AND SALTS

1. Determine whether the following substances are *ACID (A)*, *BASE (B)* or *SALT (S)*. Write the appropriate letter next to the substance.

- | | | | |
|---------------------|--------------------------|---|---------------------------------------|
| (a) NaCl | (b) MgO | (c) zinc hydroxide | (d) HNO ₃ |
| (e) copper sulphate | (f) copper oxide | (g) Al ₂ (SO ₄) ₃ | (h) H ₂ SO ₄ |
| (i) NaOH | (j) HCl | (k) FeCl ₃ | (l) Ca(NO ₃) ₂ |
| (m) sodium oxide | (n) CH ₃ COOH | (o) potassium acetate | |

2. Complete the following general word equations for acid reactions.

ACID + METAL → _____ + _____

ACID + BASE → _____ + _____

ACID + CARBONATE → _____ + _____
+ _____

3. For the following acid reactions, *complete the word equations*. Use the general equations in Q2 above to help you.

(a) magnesium oxide + sulphuric acid →

(b) aluminium + hydrochloric acid →

(c) sodium carbonate + sulphuric acid →

(d) potassium hydroxide + nitric acid →

(e) calcium + acetic acid →

(f) copper oxide + acetic acid →

(g) iron carbonate + hydrochloric acid →

(h) zinc + sulphuric acid →

(i) calcium carbonate + nitric acid →